How to Execute Task 2: A Sentence Through the Levels of NLP

This experiment uses **spaCy** to analyze a sentence at different NLP levels such as morphology, syntax, semantics, and discourse.

Step 1: Install spaCy

Open Command Prompt or Terminal and run:

```
pip install spacy
```

Step 2: Download the English Language Model

The program uses the model **en_core_web_sm**.

Run:

```
python -m spacy download en_core_web_sm
```

Step 3: Create the Python Program

Save the following code as sentence_levels.py:

```
import spacy

nlp = spacy.load("en_core_web_sm")

doc = nlp(
    "The children in Pune didn't eat "
    "the mangoes because they were raw."
)

 for t in doc:
     print(
         t.text,
         t.lemma_,
         t.morph,
         t.pos_,
         t.dep_,
         t.head.text
     )

```

```
print("\nNamed Entities:")
print(doc.ents)
```

Step 4: Run the Program

```
python sentence_levels.py
```

Sample Output

```
The the Definite=Def|PronType=Art DET det children
children child Number=Plur NOUN nsubj eat
in in ADP prep children
Pune Pune Number=Sing PROPN pobj in
did do AUX aux eat
n't not PART neg eat
eat eat VERB ROOT eat
the the DET det mangoes
mangoes mango Number=Plur NOUN dobj eat
 because because SCONJ mark were
 they they Case=Nom|Number=Plur PRON nsubj were
 were be Tense=Past VERB advcl eat
 raw raw ADJ acomp were
 . . PUNCT punct eat
```

Named Entity Output:

```
(Pune,)
```

Analysis at Each NLP Level

1. Morphological Level

Morphology studies the internal structure of words.

Word	Lemma	Morphological Features
children	child	Number=Plur
mangoes	mango	Number=Plur
were	be	Past Tense
they	they	Plural Pronoun

Output Columns Used:

```
t.lemma_
```

```
t.morph
```

Observation:

- "children" → root form is **child**
- "mangoes" → root form is **mango**
- "were" → root form is **be**

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2. Lexical Level

Lexical analysis identifies word categories.

Word	POS Tag
children	NOUN
Pune	PROPN
eat	VERB
raw	ADJ

Output Column Used:

```
t.pos_
```

---

3. Syntactic Level

Syntax studies grammatical relationships.

Word	Dependency	Head Word
children	nsubj	eat
mangoes	dobj	eat
were	advcl	eat
raw	acomp	were

Output Columns Used:

```
t.dep_
t.head.text
```

Dependency Structure

```
eat
├── children (subject)
├── mangoes (object)
└── were raw (reason clause)
```

---

4. Semantic Level

Semantics focuses on meaning.

Named Entity Recognition

Output:

```
Pune
```

Entity Type:

```
GPE (Geo-Political Entity)
```

Output Used:

```
doc.ents
```

Meaning Representation

```
EAT(children, mangoes)
CAUSE(raw(mangoes))
LOCATION(Pune)
```

Interpretation:

> The children did not eat the mangoes because the mangoes were raw.

---

5. Pragmatic Level

Pragmatics considers context.

Sentence:

```
The children in Pune didn't eat the mangoes because they were raw.
```

The listener understands that:

- The mangoes were unripe.
- Therefore the children refused to eat them.

Context helps derive the intended meaning beyond grammar.

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6. Discourse Level

Discourse concerns connections across sentences.

The sentence contains the pronoun:

```
they
```

Question:

```
Who does "they" refer to?
```

Possible interpretations:

1. The children were raw ❌
2. The mangoes were raw ✅

Humans easily infer that **they = mangoes** because mangoes can be raw.

spaCy identifies grammar and dependencies but does **not fully resolve this coreference relationship** in this simple example.

Exercise Answers

Label each column with its NLP level

Column	NLP Layer
lemma_	Morphology
morph	Morphology
pos_	Lexical / Syntax
dep_	Syntax
head.text	Syntax
doc.ents	Semantics

Which word does "they" refer to?

Answer: mangoes

Reason:

```
The children in Pune didn't eat the mangoes because they were raw.
```

The adjective **raw** naturally describes **mangoes**, not **children**.

Conclusion

This experiment demonstrates how a single sentence can be analyzed across multiple NLP layers:

NLP Layer	Information Extracted
Morphology	Lemmas and grammatical features
Lexical	Parts of speech
Syntax	Dependency relations
Semantics	Named entities and meaning
Pragmatics	Context-based interpretation
Discourse	Pronoun reference and coherence

The experiment shows that NLP understanding progresses from individual word forms to complete interpretation of meaning and context.